Linbo Tang

+1 608-886-7491 | linbo0204@outlook.com | LinkedIn | GitHub | Portfolio

Education

Harvard University — MA, USA
S.M in Data Science; ***GPA: 3.9/4.0*** — *Sep 2024 – May 2026*

University of Wisconsin - Madison — WI, USA
B.S in Computer Science and Data Science; ***GPA: 4.0/4.0*** — *Sep 2020 – Dec 2023*

Publications

DO DEEP STEREO MODELS LEARN SHAPE PRIORS?
Linbo Tang, Aneel Damaraju, Todd Zickler
NeurIPS 2026 Workshop on Physical World Models

SCANNER-ADAPTIVE COIL-LEVEL DENOISING FOR DIFFUSION MRI USING EXPLICIT NOISE PRIORS
Linbo Tang, Qiang Liu, Lipeng Ning, Yogesh Rathi
ISBI 2026 (Oral)

Work Experience

Kempner Institute, Harvard University — Sep 2026 – Present
AI Fellow

- Investigate **multimodal generative models** for geometric and physical understanding under ambiguous or incomplete observations.
- Explore **3D motion representations** to support physical reasoning and planning in simulated environments.
- Develop and evaluate **agentic workflows** that use visual and motion-based feedback to refine generated scene dynamics.

Research Experience

Psychiatry and Radiology Image Computing Group — Sep 2024 – May 2026
Harvard Medical School — *Advisor: Prof. Yogesh Rathi*

- Developed a novel **scanner-adaptive, coil-level** denoising framework that conditions the network on measured noise statistics to achieve robust performance across scanners and protocols, accepcted **ISBI 2026**
- Introduced an attention-level conditioning mechanism via feature-wise modulation to dynamically adjust denoising strength according to local noise context.
- Evaluated the framework on in-vivo multi-coil diffusion MRI data, achieving up to a **23%** reduction in residual noise compared to the next best-performing method

Zickler's Vision Lab — Dec 2024 – Present
Harvard University — *Advisor: Prof. Todd Zickler*

- Developed a **diffusion-based stereo reasoning model** with an hourglass transformer to jointly predict disparity and geometry fields for improved 3D scene understanding.
- Targeted challenging stereo settings including **occlusion, weak texture, camouflage, and ambiguous visual correspondences**, where conventional matching-based methods often fail.
- Built a large-scale **synthetic physical-scene pipeline** generating rectified stereo pairs and geometric labels for controlled training, ablation, and evaluation of geometry-aware reasoning models.

Metadata-Conditioned MRI Reconstruction — Apr 2025 – Apr 2026
MIT CSAIL — *Advisor: Dr. Giannis Daras*

- Developed **ContextMRI**, a diffusion-based MRI reconstruction framework that leverages **fine-grained metadata embeddings** to condition on imaging parameters such as TR, TE, and flip angle.
- Designed a **structured embedding architecture** separating categorical and numerical attributes, using discrete embeddings and sinusoidal encodings respectively for improved quantitative representation.

- Introduced an **out-of-distribution (OOD) training strategy** by incorporating cardiac MRI data with distinct parameter ranges, forcing reliance on metadata priors and improving generalization.

Flow Matching for RL-Finetuned Robot Policies Jan 2026 – Present

Harvard University *Advisor: Prof. Stephanie Gil*

- Extended DSRL by replacing diffusion policy backbones with **flow matching**, enabling faster action generation and more efficient inference for real-time robot manipulation.
- Evaluated flow-matching-based policies on **MuJoCo** and **LIBERO** benchmarks to study policy performance, inference efficiency, and generalization across manipulation tasks.
- Integrated **vision and tactile sensing** as multimodal observations for contact-aware policy learning, enabling reasoning over physical structure in dexterous manipulation tasks within IsaacSim.

MSC Computer Vision Group Jan 2023 – Feb 2024

University of Wisconsin - Madison *Advisor: Prof. Vikas Singh*

- Curating **EmotionNet5k**, a novel selfie video dataset containing 5000 samples of facial expressions, with over 300 stars on GitHub;
- Implementing a SOTA learnable wavelet model to remove co-founders and achieve a **3.13%** increase in accuracy on the downstream Facial Emotion Recognition (FER) task.

Internship

Ubisoft La Forge Apr 2024 – Aug 2024

Research Student; La Forge *Advisor: Alexis Rolland*

- Developed a generative base-material creation pipeline utilizing **StableDiffusion XL** and **inverse rendering** models to obtain rendered textures from text prompts and physically based rendering (PBR) maps from rendered textures.
- Refined a SwinIR-based super-resolution model for PBR maps by introducing a multi-branch architecture that separately processes base color, normal, roughness, and height maps while enforcing inter-map consistency.
- Achieved higher perceptual fidelity and improved structural coherence across PBR maps compared to standard single-branch and texture-domain super-resolution baselines.

Teach Experience

Teaching Fellow Aug 2025 – May 2026

Harvard University *Advisor: Prof. Michael Brenner*

- Led weekly discussion sections for **AM 215: Mathematical Modeling**, covering topics such as dimensional analysis, Monte-Carlo simulation, and extreme-value statistics.
- Collaborated with the teaching staff on the design of homework assignments and course structure, providing individualized guidance to students throughout the semester.

Undergraduate Teaching Assistant Sep 2021 – May 2023

University of Wisconsin–Madison *Advisor: Prof. Frederic Sala*

- Assisted in teaching four core courses: CS 200 (Programming I), CS 300 (Programming II), CS 400 (Programming III), and CS 540 (Introduction to Artificial Intelligence).
- Guided students through fundamental and intermediate programming concepts, data structures, and AI algorithms, emphasizing debugging and problem-solving strategies.
- Conducted tutoring sessions twice weekly, supporting over **400 students** across four semesters.
- Created and maintained a public teaching portfolio showcasing section materials, code examples, and instructional resources.

Technical Skills

Programming Languages: Java, Python, C/C++, R, JavaScript, CSS, HTML, Kotlin;

Frameworks: Pytorch, Tensorflow, Scikit-Learn, MySQL, OpenCV, Flask, React, Selenium;

Platforms: Isaac Sim, UE4, AWS, GCP, Docker, Linux, Git.